ORGNZ-06: Security Context

> **Series:** ORGNZ — Organize Data: Buckets, Segments, Security | **Notebook:** 6 of 10 | **Created:** January 2026 | **Last Updated:** 02/19/2026

Overview

If permissions on deployment-level attributes or bucket level are insufficient, Dynatrace allows you to set up **fine-grained permissions** by adding a `dt.security_context` attribute to your data. This enables record-level access control that scales beyond bucket limits.

Prerequisites

Requirement	Details
Dynatrace Environment	SaaS environment with Grail enabled
Permissions	`storage:logs:read`, OpenPipeline configuration access
Knowledge	Completed ORGNZ-05 (Bucket-Level Access Control)
Data	Logs with `dt.security_context` field (or ability to configure)

Table of Contents

-------------|----------|
Dynatrace Account	Account-level administrative access
Permissions	OpenPipeline configuration, IAM policy management
Knowledge	Completed ORGNZ-04 and ORGNZ-05

Learning Objectives

By the end of this notebook, you will:
- Understand the purpose and structure of security context
- Know how to assign security context via OpenPipeline
- Create IAM policies that use security context
- Design hierarchical security context patterns

What is Security Context?

Security context is a reserved field (`dt.security_context`) that enables custom authorization:

Aspect	Description
Field name	`dt.security_context`
Type	String or array of strings
Purpose	Custom authorization beyond standard attributes
Source	OpenPipeline, OneAgent, Kubernetes labels
Enforcement	IAM policies filter at query time

Why Use Security Context?

Scenario	Without Security Context	With Security Context
100 teams sharing data	Need 100 buckets	Single bucket, 100 context values
Dynamic team membership	Static bucket assignment	Update context via tags
Hierarchical access	Complex policy combinations	Hierarchical string encoding

Security Context Values

Simple Values

```
dt.security_context: "team-a"
dt.security_context: "finance"
dt.security_context: "sec-lvl-7"
```

Hierarchical Encoding

Encode hierarchy in a string for flexible access patterns:

```
dt.security_context: "department-A/team-1/project-x"
dt.security_context: "org1/finance/compliance"
dt.security_context: "region-us/aws-account-123/team-platform"
```

Array Values

Records can have multiple security context values:

```
dt.security_context: ["team-a", "project-x", "compliance"]
```

Setting Security Context

Via OpenPipeline

Configure OpenPipeline to add security context based on record attributes:

1. Go to **Settings** > **Log Processing** > **OpenPipeline**
2. Select your pipeline
3. Go to **Permission** tab
4. Add **Set Security Context** processor

```
# OpenPipeline security context processor
processors:
  - type: security-context
    rules:
      - condition: "host.group starts-with 'finance-'"
        context: "lob:finance"
      - condition: "k8s.namespace.name == 'checkout'"
        context: "team:checkout"
      - condition: "matchesValue(service.name, 'payment-*')"
        context: "team:payments"
```

Via OneAgent

OneAgent can set security context based on:

Source	Configuration
Host groups	Automatic from host group membership
Kubernetes labels	Map labels to security context
Cloud metadata	Use cloud account/project info
Custom tags	Reference existing tags

Via Kubernetes

Use labels or annotations to set security context:

```
metadata:
  labels:
    dynatrace.com/security-context: "team:checkout"
  annotations:
    dynatrace.com/security-context: "compliance:pci"
```

IAM Policies with Security Context

Basic Security Context Policy

```
{
  "name": "team-a-data-access",
  "description": "Team A can access data with team-a security context",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context = 'team-a';",
  "tags": ["team:team-a"]
}
```

Hierarchical Access with MATCH

Grant access to entire department:

```
{
  "name": "finance-department-access",
  "description": "Finance department can access all finance sub-teams",
  "statementQuery": "ALLOW storage:buckets:read WHERE storage:bucket-name
STARTSWITH 'default_'; ALLOW storage:logs:read WHERE
storage:dt.security_context MATCH ('finance/*');",
  "tags": ["department:finance"]
}
```

Array Field Access with MATCH

> **Important**: When `dt.security_context` holds an array, you MUST use `MATCH` operator. Using `=`, `STARTSWITH`, or `IN` will always return false for array fields.

```
// Correct - MATCH for array fields
ALLOW storage:logs:read WHERE storage:dt.security_context MATCH ("crn-70400-*");

// Wrong - = won't work for arrays
ALLOW storage:logs:read WHERE storage:dt.security_context = "crn-70400-team";
```

Querying Security Context

> **Lab Exercise:** Complete Exercises 1-2 in **ORGNZ-06 LAB** for hands-on practice with these concepts.

Security Context Patterns

Pattern 1: Team-Based Access

```
Context value: "team:&lt;team-name&gt;"
Examples: "team:platform", "team:checkout", "team:payments"
Policy: ALLOW ... WHERE storage:dt.security_context = "team:platform"
```

Pattern 2: Line of Business

```
Context value: "lob:&lt;business-unit&gt;"
Examples: "lob:finance", "lob:retail", "lob:healthcare"
Policy: ALLOW ... WHERE storage:dt.security_context = "lob:finance"
```

Pattern 3: Hierarchical Organization

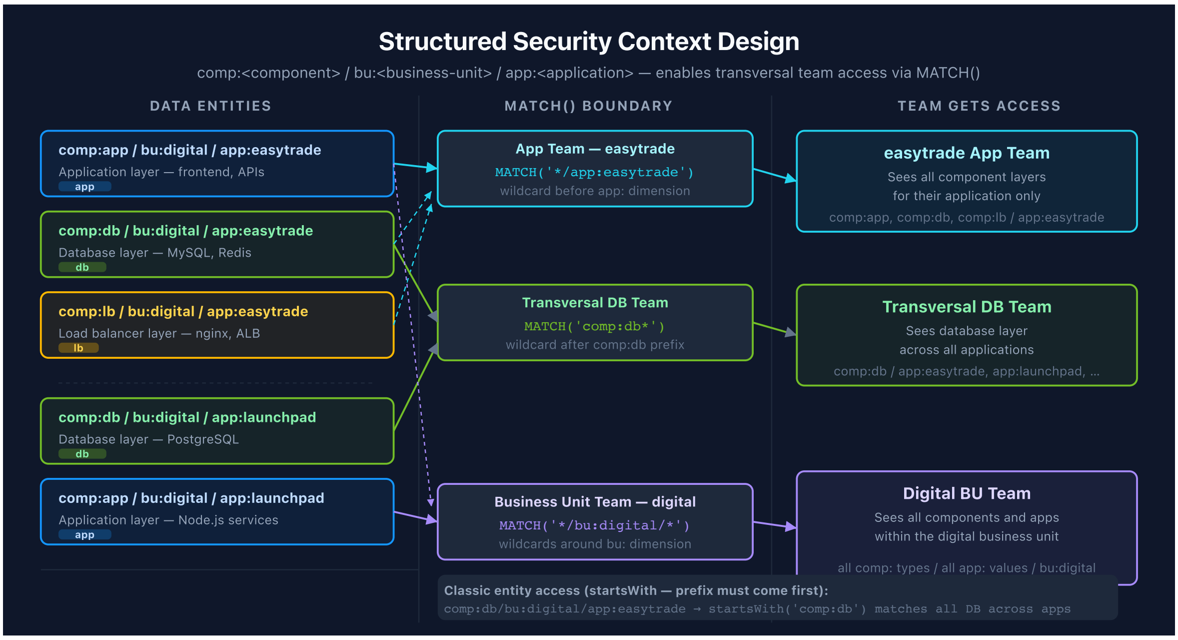
```
Context value: "&lt;org&gt;/&lt;department&gt;/&lt;team&gt;"
Examples: "acme/engineering/platform", "acme/finance/compliance"

Department-level access:
ALLOW ... WHERE storage:dt.security_context MATCH ("acme/engineering/*")

Team-level access:
ALLOW ... WHERE storage:dt.security_context = "acme/engineering/platform"
```

Pattern 4: Cloud Account Mapping

```
Context value: "&lt;cloud&gt;/&lt;account&gt;"
Examples: "aws/123456789012", "gcp/my-project-id"
Policy: ALLOW ... WHERE storage:dt.security_context MATCH ("aws/*")
```

Pattern 5: Multi-Dimensional Structured Context

Encode multiple dimensions into a single string for transversal team access — teams that need cross-application visibility (database, networking, OS) without being tied to a specific application.



```
Context format: "comp:&lt;component&gt;/bu:&lt;business-
unit&gt;/app:&lt;application&gt;"
Examples:
  "comp:app/bu:digital/app:easytrade"  — application layer, digital BU
  "comp:db/bu:digital/app:easytrade"   — database layer, digital BU
  "comp:lb/bu:digital/app:easytrade"   — load balancer layer, digital BU
```

MATCH() access patterns (Grail / 3rd Gen storage domain):

Team Type	MATCH() Expression	Grants Access To
App team	`MATCH('*/app:easytrade')`	All components for one application
Transversal DB team	`MATCH('comp:db*')`	Database layer across all applications
Business unit team	`MATCH('*/bu:digital/*')`	All components and apps within digital BU

startsWith patterns (Classic entity access):

```
// App team — enumerate each component explicitly
ALLOW storage:logs:read WHERE storage:dt.security_context startsWith
"comp:app/bu:digital/app:easytrade"
ALLOW storage:logs:read WHERE storage:dt.security_context startsWith
"comp:db/bu:digital/app:easytrade"

// Transversal DB team — one prefix covers all apps
ALLOW storage:logs:read WHERE storage:dt.security_context startsWith
"comp:db"
```

> **Dimension order matters for Classic entities.** Place the dimension used for transversal access *first* (typically `comp`). For the full two-domain comparison and multi-value context roadmap, see **IAM-04: Policy Authoring** and **IAM-05: Boundary Design**.

Security Context Best Practices

Practice	Rationale
Use consistent naming conventions	Enables pattern matching in policies
Plan hierarchy before implementation	Hard to change after data is ingested
Use prefixes for context types	`team:`, `lob:`, `env:` for clarity
Document context values	Governance and audit requirements
Test with sample data	Verify context is assigned correctly
Use MATCH for array fields	Required for correct policy evaluation
Encode multiple dimensions for transversal access	`comp/bu/app` format scales to cross-application teams without per-app policies

Entity Security Context

Security context can also be set on entities (not just records):

1. Go to **Settings** > **Topology model** > **Grail Security Context**
2. Define rules to assign security context to entities
3. Entities inherit context to related records

This enables entity-level access control for services, hosts, and other monitored entities.

Next Steps

Continue with the ORGNZ series:
- **ORGNZ-07**: Advanced Permission Patterns

References

- Configure advanced permissions with security context
- Set up Grail permissions for OneAgent
- Grant access to entities with security context

This notebook was AI-generated from Dynatrace documentation and enterprise best practices. It is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.